

Design Project 3

1. Considering the foundation block supporting the bridge central pier (i.e. 20 m length, 10 m wide and 8 meters high), calculate/demonstrate: a) The thermal stresses that might arise through the use of a single pouring lift. Please consider: o Foundation on rock: $E_{rock} = 40000 \text{ MPa}$.

- Data given:
 - $L=20\text{ M}$
 - $W=10\text{ M}$
 - $H=8\text{ M}$
 - $E_{\text{rock}}= 40,000\text{ MPa}$
- For Thermal Stresses :
- We are using the mix from Design project 2.
 - 28-day compressive strength of concrete = 37.8 MPa
 - Binder (cement + fly ash + silica fume) = 405.94 Kg/m^3
 - Fine aggregates = 718.08 Kg
 - Coarse aggregates = 1012 Kg/m^3
 - Water = 73.4 Kg

Materials	m - content (kg/m³)	c - Specific heat (kcal/kg*°C)	q= m*c(kcal/m³*°C)	Temperature (C)		Δt	Q= q*Δt (kcal/C)
				Ti	Tf		
Cement	406	0.2	81.2	22	0	22	1786.4
Sand	718.08	0.19	136.4352	22	0	22	3001.5744
Course Aggregate	1012	0.24	242.88	22	0	22	5343.36
Water	0	1	0	12	0	12	0
Humidity Sand	43.08	1	43.08	22	0	22	947.76
Humidity course	20.24	1	20.24	22	0	22	445.28
Ice	73.4	0.5	36.7	-10	0	-10	-367
Ice fusion	73.4	0	0	0	0	0	-5872
Ice-water	73.4	1	73.4	0	20	-20	-1468
Q mix	1	1000	1000	1	0	1	1000
		q summation	633.9352			Q summation	4817.3744
				Temperature at mixture			7.6
				Temperature Rise(Transportation-C)			2
				Placing Temperature			9.6

- For the adiabatic temperature rise:

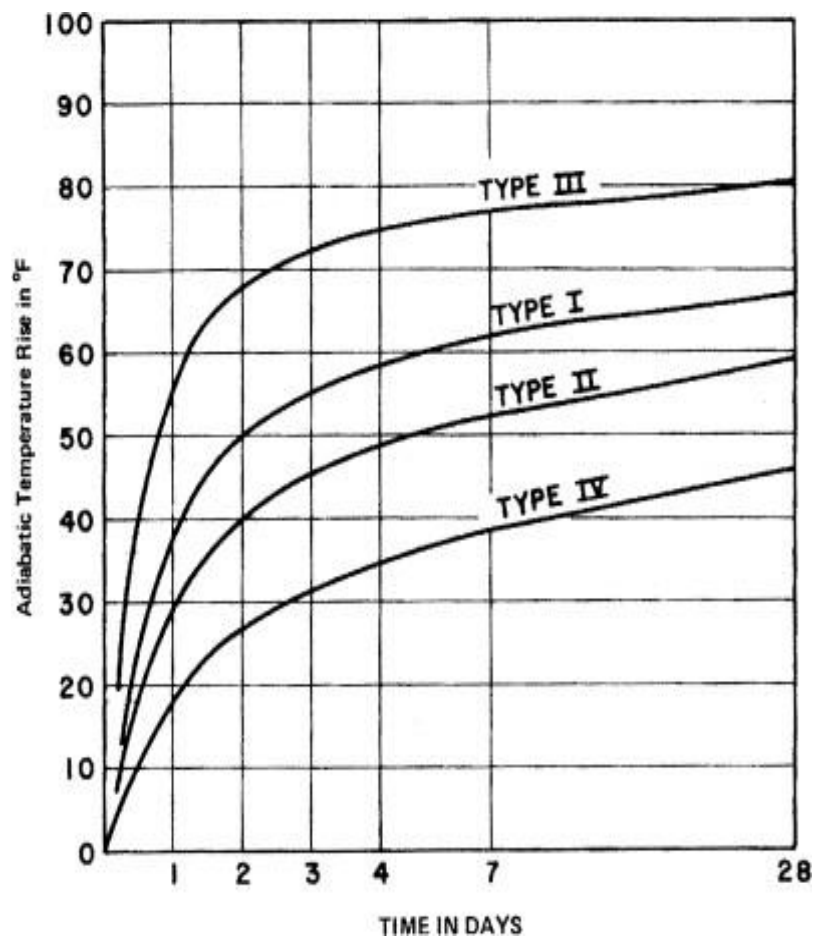
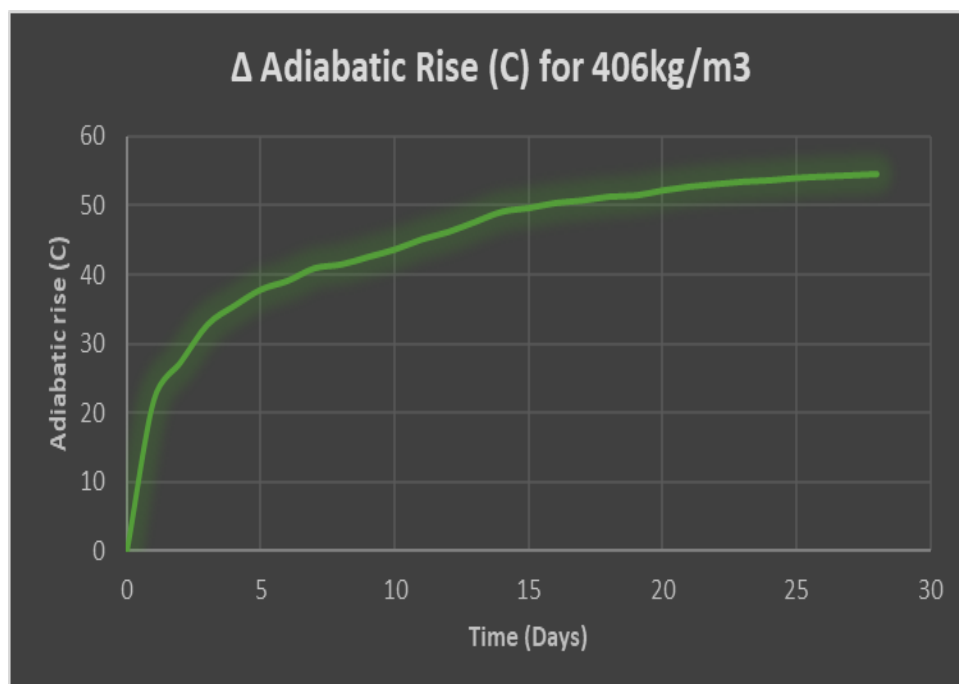


fig 1 ACI 207 2R-95

- As per the above graph and literature of ACI 207 2R-95, we have adiabatic temperature rise for 223kg/m^3 (type 1) which is around 70F and 22C. So, we can interpolate and get the temperature rise for 406kg/m^3 .

Time (days)	Adiabatic rise (C) for 223kg/m ³	Δ Adiabatic Rise (C) for 406kg/m ³	Δ Adiabatic Rise (C) for 406kg/m ³
0	0	0	0
1	9	16.39	16.39
2	11	20.03	3.64
3	13.2	24.03	4.01
4	13.9	25.31	1.27
5	14.8	26.95	1.64
6	15.7	28.58	1.64
7	16.5	30.04	1.46
8	16.9	30.77	0.73
9	17.4	31.68	0.91
10	17.8	32.41	0.73
11	18	32.77	0.36
12	18.35	33.41	0.64
13	18.8	34.23	0.82
14	19.5	35.50	1.27
15	20	36.41	0.91
16	20.3	36.96	0.55
17	20.5	37.32	0.36
18	20.8	37.87	0.55
19	20.9	38.05	0.18
20	21	38.23	0.18
21	21.3	38.78	0.55
22	21.45	39.05	0.27
23	21.6	39.33	0.27
24	21.7	39.51	0.18
25	21.8	39.69	0.18
26	21.9	39.87	0.18
27	21.95	39.96	0.09
28	22	40.05	0.09

➤ Adiabatic Temperature rise:



➤ Calculation of only single lift over 28 days:

➤ Temperature Rise:

- Here, I am only attaching 2 day calculations.
- The adiabatic temperature rise for day 0 is 0C and 1st day 16.4C.
- Temperature increase for 1-day will be
 $(0+16.4/2) = 8.2$ at 0.2 m spacing interval.
- For more the temperature rise is as below.

structure	Concrete Lift	Time (days)	0	1	2
		Height	Δ Temp (C)	16.386	Δ Temp (C) 3.641
Concrete Block	Lift	8	0.000	0.000	0.000
		7.8	0	16.386	8.19282511
		7.6	0	16.386	16.3856502
		7.4	0	16.386	16.3856502
		7.2	0	16.386	16.3856502
		7	0	16.386	16.3856502
		6.8	0	16.386	16.3856502
		6.6	0	16.386	16.3856502
		6.4	0	16.386	16.3856502
		6.2	0	16.386	16.3856502
		6	0	16.386	16.3856502
		5.8	0	16.386	16.3856502
		5.6	0	16.386	16.3856502
		5.4	0	16.386	16.3856502
		5.2	0	16.386	16.3856502
		5	0	16.386	16.3856502
		4.8	0	16.386	16.3856502
		4.6	0	16.386	16.3856502
		4.4	0	16.386	16.3856502
		4.2	0	16.386	16.3856502
		4	0	16.386	16.3856502
		3.8	0	16.386	16.3856502
		3.6	0	16.386	16.3856502
		3.4	0	16.386	16.3856502
		3.2	0	16.386	16.3856502
		3	0	16.386	16.3856502
		2.8	0	16.386	16.3856502
		2.6	0	16.386	16.3856502
		2.4	0	16.386	16.3856502
		2.2	0	16.386	16.3856502
		2	0	16.386	16.3856502
		1.8	0	16.386	16.3856502
		1.6	0	16.386	16.3856502
		1.4	0	16.386	16.3856502
		1.2	0	16.386	16.3856502
		1	0	16.386	16.3856502
		0.8	0	16.386	16.3856502
		0.6	0	16.386	16.3856502
		0.4	0	16.386	12.2892377
		0.2	0	8.192825112	8.19282511
Soil	Soil	-0.2	0	0	4.09641256
		-0.4	0	0	0
		-0.6	0	0	0
		-0.8	0	0	0
		-1	0	0	0
		-1.2	0	0	0
		-1.4	0	0	0
		-1.6	0	0	0
		-1.8	0	0	0
		-2	0	0	0
		-2.2	0	0	0
		-2.4	0	0	0
		-2.6	0	0	0
		-2.8	0	0	0
		-3	0	0	0
		-3.2	0	0	0
		-3.4	0	0	0
		-3.6	0	0	0
		-3.8	0	0	0
		-4	0	0	0
		-4.2	0	0	0
		-4.4	0	0	0
		-4.6	0	0	0
		-4.8	0	0	0
		-5	0	0	0
		-5.2	0	0	0
		-5.4	0	0	0
		-5.6	0	0	0
		-5.8	0	0	0
		-6	0	0	0
		-6.2	0	0	0
		-6.4	0	0	0
		-6.6	0	0	0
		-6.8	0	0	0
		-7	0	0	0
		-7.2	0	0	0
		-7.4	0	0	0
		-7.6	0	0	0
		-7.8	0	0	0
		-8	0	0	0

- Temperature difference between 1st day and 2nd day is 3.6C at 0.2 m spacing interval, and temperature rise is $(0+16.4/2) = 8.2C$.

- As a result, the temperature for 1 day will be $= 8.2+3.6/2 = 10.01C$.

Temperature difference between 1st day and 2nd day is 3.6C at 0.4 m spacing interval, and temperature rise is $[(16.4+8.2)/2] = 12.3C$.

- As a result, the temperature for 1 day will be $= 12.3+3.64 = 15.93C$

Temperature difference at 0.6m spacing interval the rise is $[(16.4+16.4)/2] = 16.4C$.

- As a result, the temperature for 1 day will be $= 16.4+3.64 = 20.02C$

- Same way the calculation is done till 7.6m height.

Temperature difference at 7.8m spacing interval the rise is $[(0+16.4)/2] = 8.2C$

- As a result, the temperature for 1 day will be $= 8.2+3.64 = 11.83C$

➤ Stress :

- The construction is done over the fall so the average temperature in fall in Ottawa is around 20C (from the internet). So, we have taken the ambient temperature of 18C.
- If we do not take extra measures to control the temperature, the temperature of the concrete in the forms will generally be higher than the ambient air temperature.
- The placing temperature is calculated as above which is 9.6C.
- For the kr: L/H

$$a. \geq 2.5 \rightarrow Kr = \frac{\frac{L}{H}-2}{\frac{L}{H}+1}$$

$$b. < 2.5 \rightarrow Kr = \frac{\frac{L}{H}-1}{\frac{L}{H}+10}$$

- For the block, $L/H = 2.5$
- We are also using the plastic sheet to reduce the thermal stress.
- Hence, $Kr = (2.5 - 2)/(2.5+1)/2 = 0.143/2 = 0.0715$ (divided by 2 because plastic sheet used)

- Now, for the Type/stiffness of the soil: K_f

$$K_f = \frac{1}{1 + \frac{A_c * E_c}{A_f * E_f}}$$

$$A_c = 20 * 10 = 200 \text{ m}^2 \quad A_f = 2.5 * 200 = 500 \text{ m}^2$$

$$E_f = 40000 \text{ MPa} \quad A_f = 2.5 * 200 = 500 \text{ m}^2$$

$$E_c = 30220.3 \text{ MPa} \quad E_f = 40000 \text{ MPa}$$

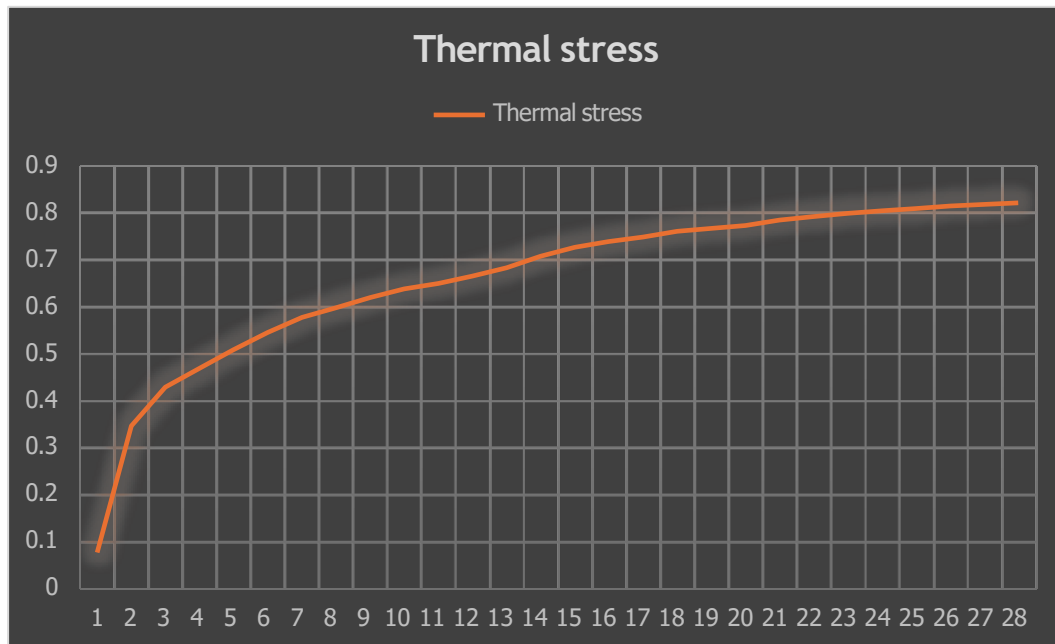
$$\text{As a result, } K_f = 1/1 + [(200 * 30220.3)/(500 * 40000)] = 0.77$$

$$\text{Thus, } K_{total} = 0.0715 * 0.77 = 0.055$$

- So the calculation of stress of single lift is as below:
- $\sigma = E * \alpha * \Delta T * Kr * K_f * \text{creep}$
- At the 0.4m $\sigma = (0.00001 * 0.0714 * 0.767 * 1 * (18 - 9.6) * 17547.7) = 0.077$

structure	Concrete Lift	Time (days)	0	1	2	
		Height	Δ Temp (C)	16.386	Δ Temp (C)	3.641
Concrete Block	I lift	8		-0.080860551		0.112435924
		7.8		0.076856125		0.251049712
		7.6		0.076856125		0.347013103
		7.4		0.076856125		0.347013103
		7.2		0.076856125		0.347013103
		7		0.076856125		0.347013103
		6.8		0.076856125		0.347013103
		6.6		0.076856125		0.347013103
		6.4		0.076856125		0.347013103
		6.2		0.076856125		0.347013103
		6		0.076856125		0.347013103
		5.8		0.076856125		0.347013103
		5.6		0.076856125		0.347013103
		5.4		0.076856125		0.347013103
		5.2		0.076856125		0.347013103
		5		0.076856125		0.347013103
		4.8		0.076856125		0.347013103
		4.6		0.076856125		0.347013103
		4.4		0.076856125		0.347013103
		4.2		0.076856125		0.347013103
		4		0.076856125		0.347013103
		3.8		0.076856125		0.347013103
		3.6		0.076856125		0.347013103
		3.4		0.076856125		0.347013103
		3.2		0.076856125		0.347013103
		3		0.076856125		0.347013103
		2.8		0.076856125		0.347013103
		2.6		0.076856125		0.347013103
		2.4		0.076856125		0.347013103
		2.2		0.076856125		0.347013103
		2		0.076856125		0.347013103
		1.8		0.076856125		0.347013103
		1.6		0.076856125		0.347013103
		1.4		0.076856125		0.347013103
		1.2		0.076856125		0.347013103
		1		0.076856125		0.347013103
		0.8		0.076856125		0.347013103
		0.6		0.076856125		0.347013103
		0.4		0.076856125		0.299031408
		0.2		-0.002002213		0.229724513
		-0.2		-0.080860551		0.160417619
Soil	Soil	-0.4				0.112435924
	Soil	-0.6				0.112435924
	Soil	-0.8				0.112435924
	Soil	-1				0.112435924
	Soil	-1.2				0.112435924
	Soil	-1.4				
	Soil	-1.6				
	Soil	-1.8				
	Soil	-2				
	Soil	-2.2				
	Soil	-2.4				
	Soil	-2.6				
	Soil	-2.8				
	Soil	-3				
	Soil	-3.2				
	Soil	-3.4				
	Soil	-3.6				
	Soil	-3.8				
	Soil	-4				
	Soil	-4.2				
	Soil	-4.4				
	Soil	-4.6				
	Soil	-4.8				
	Soil	-5				
	Soil	-5.2				
	Soil	-5.4				
	Soil	-5.6				
	Soil	-5.8				
	Soil	-6				
	Soil	-6.2				
	Soil	-6.4				
	Soil	-6.6				
	Soil	-6.8				
	Soil	-7				
	Soil	-7.2				
	Soil	-7.4				
	Soil	-7.6				
	Soil	-7.8				
	Soil	-8				

- After 28 days the maximum thermal stress is 0.82 which is low because we added plastic sheet. So, even in the single the concrete will not have thermal cracking as the thermal stresses is low.



2. A proposed performance-based design (PBD) solution (i.e. placing temperature, selection of materials - including ice, number of lifts and reinforcement, if needed) to not have problems related to thermal stresses. Calculations should be done over the first month after pouring (i.e., 28 days).

➤ Performance based design:

- For performance based design, to not have any problem related to thermal stresses we have get placing temperature as low as possible, so we got placing temperature of 9.6C.
- We have also provided 1 day break and divided the pouring in two lifts (i.e 4 m each) to not have problem related thermal cracking.
- We have also provided Plastic sheet to reduce the K_r which reduce thermal stress so we will not have problems about thermal cracking.

I. **Temperature rise for two lifts for 28 days:**

- Example of the temperature rise in first and second lift is as below:
- The temperature rise at the 3rd day;
 At 4 m $(5.9 + 4/2 + 16.4/2) = 16.11^\circ\text{C}$
 At 4.2m $(0 + 16.4) = 16.4^\circ\text{C}$
- The temperature rise in the 4th day:
 At 4 m $(15.2 + 1.2/2 + 3.64/2) = 17.66^\circ\text{C}$
 At 4.2m $(16.24 + 3.64) = 19.89^\circ\text{C}$
- So, the same calculation is done until 8m.

strcuture	Hieght	Concrete Lift	Time (days)	0	1	2	3		4		
			Height	Δ Temp (C)	16.386	Δ Temp (C)	3.641	Δ Temp (C) (1st)	4.005	Δ Temp (C) (1st)	1.274
				Δ Temp (C)	16.386	Δ Temp (C)	3.641	Δ Temp (C) (2nd)	16.386	Δ Temp (C) (2nd)	3.641
Concrete Block	8	Second Lift	8	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	7.8					0	0	16.386	8.193	11.834	
	7.6					0	0	16.386	16.386	20.027	
	7.4					0	0	16.386	16.386	20.027	
	7.2					0	0	16.386	16.386	20.027	
	7					0	0	16.386	16.386	20.027	
	6.8					0	0	16.386	16.386	20.027	
	6.6					0	0	16.386	16.386	20.027	
	6.4					0	0	16.386	16.386	20.027	
	6.2					0	0	16.386	16.386	20.027	
	6					0	0	16.386	16.386	20.027	
	5.8					0	0	16.386	16.386	20.027	
	5.6					0	0	16.386	16.386	20.027	
	5.4					0	0	16.386	16.386	20.027	
	5.2					0	0	16.386	16.386	20.027	
	5					0	0	16.386	16.386	20.027	
	4.8					0	0	16.386	16.386	20.027	
	4.6					0	0	16.386	16.386	20.027	
	4.4					0	0	16.386	16.386	20.027	
	4.2					0	0	16.386	16.249	19.890	
	4	First Lift	4	0	0	0	0	5.917040359	16.11256	15.202	17.66009
	3.8			0	16.386	8.19282511	11.834	10.01345291	14.019	18.024	19.299
	3.6			0	16.386	16.3856502	20.027	15.93049327	19.936	19.026	20.300
	3.4			0	16.386	16.3856502	20.027	20.02690583	24.032	21.984	23.259
	3.2			0	16.386	16.3856502	20.027	20.02690583	24.032	24.032	25.307
	3			0	16.386	16.3856502	20.027	20.02690583	24.032	24.032	25.307
	2.8			0	16.386	16.3856502	20.027	20.02690583	24.032	24.032	25.307
	2.6			0	16.386	16.3856502	20.027	20.02690583	24.032	24.032	25.307
	2.4			0	16.386	16.3856502	20.027	20.02690583	24.032	24.032	25.307
	2.2			0	16.386	16.3856502	20.027	20.02690583	24.032	24.032	25.307
	2			0	16.386	16.3856502	20.027	20.02690583	24.032	24.032	25.307
	1.8			0	16.386	16.3856502	20.027	20.02690583	24.032	24.032	25.307
	1.6			0	16.386	16.3856502	20.027	20.02690583	24.032	24.032	25.307
	1.4			0	16.386	16.3856502	20.027	20.02690583	24.032	24.032	25.307
	1.2			0	16.386	16.3856502	20.027	20.02690583	24.032	24.032	25.307
	1			0	16.386	16.3856502	20.027	20.02690583	24.032	24.032	25.307
	0.8			0	16.386	16.3856502	20.027	20.02690583	24.032	23.008	24.283
	0.6			0	16.386	16.3856502	20.027	17.97869955	21.984	21.529	22.803
	0.4			0	16.386	12.2892377	15.930	15.02017937	19.026	17.000	18.275
	0.2			0	8.192825112	8.19282511	10.01345291	10.01345291	12.01614	12.016	12.65336
Soil	0.2	Soil	-0.2	0	4.09641256	4.096412556	5.006726457	5.006726	7.032	7.032	
	0.4	Soil	-0.4	0	0	0	2.048206278	2.048206	2.503	2.503	
	0.6	Soil	-0.6	0	0	0	0	0	1.024	1.024	
	0.8	Soil	-0.8	0	0	0	0	0	0.000	0.000	
	1	Soil	-1	0	0	0	0	0	0.000	0.000	
	1.2	Soil	-1.2	0	0	0	0	0	0.000	0.000	
	1.4	Soil	-1.4	0	0	0	0	0	0.000	0.000	
	1.6	Soil	-1.6	0	0	0	0	0	0.000	0.000	
	1.8	Soil	-1.8	0	0	0	0	0	0.000	0.000	
	2	Soil	-2	0	0	0	0	0	0.000	0.000	
	2.2	Soil	-2.2	0	0	0	0	0	0.000	0.000	
	2.4	Soil	-2.4	0	0	0	0	0	0.000	0.000	
	2.6	Soil	-2.6	0	0	0	0	0	0.000	0.000	
	2.8	Soil	-2.8	0	0	0	0	0	0.000	0.000	
	3	Soil	-3	0	0	0	0	0	0.000	0.000	
	3.2	Soil	-3.2	0	0	0	0	0	0.000	0.000	
	3.4	Soil	-3.4	0	0	0	0	0	0.000	0.000	
	3.6	Soil	-3.6	0	0	0	0	0	0.000	0.000	
	3.8	Soil	-3.8	0	0	0	0	0	0.000	0.000	
	4	Soil	-4	0	0	0	0	0	0.000	0.000	
	4.2	Soil	-4.2	0	0	0	0	0	0.000	0.000	
	4.4	Soil	-4.4	0	0	0	0	0	0.000	0.000	
	4.6	Soil	-4.6	0	0	0	0	0	0.000	0.000	
	4.8	Soil	-4.8	0	0	0	0	0	0.000	0.000	
	5	Soil	-5	0	0	0	0	0	0.000	0.000	
	5.2	Soil	-5.2	0	0	0	0	0	0.000	0.000	
	5.4	Soil	-5.4	0	0	0	0	0	0.000	0.000	
	5.6	Soil	-5.6	0	0	0	0	0	0.000	0.000	
	5.8	Soil	-5.8	0	0	0	0	0	0.000	0.000	
	6	Soil	-6	0	0	0	0	0	0.000	0.000	
	6.2	Soil	-6.2	0	0	0	0	0	0.000	0.000	
	6.4	Soil	-6.4	0	0	0	0	0	0.000	0.000	
	6.6	Soil	-6.6	0	0	0	0	0	0.000	0.000	
	6.8	Soil	-6.8	0	0	0	0	0	0.000	0.000	
	7	Soil	-7	0	0	0	0	0	0.000	0.000	
	7.2	Soil	-7.2	0	0	0	0	0	0.000	0.000	
	7.4	Soil	-7.4	0	0	0	0	0	0.000	0.000	
	7.6	Soil	-7.6	0	0	0	0	0	0.000	0.000	
	7.8	Soil	-7.8	0	0	0	0	0	0.000	0.000	
	8	Soil	-8	0	0	0	0	0	0.000	0.000	

II. Stress calculation:

- The placing temperature is calculated as above which is 9.6C.
- For the kr: L/H

$$c. \geq 2.5 \rightarrow Kr = \frac{\frac{L}{H}-2}{\frac{L}{H}+1}$$

$$d. < 2.5 \rightarrow Kr = \frac{\frac{L}{H}-1}{\frac{L}{H}+10}$$

- For the block, L/H = 5
- We are also using the plastic sheet to reduce the Kr and the thermal stress also gets reduced.
- Hence, Kr = (5 - 2)/(5+1)/2 = 0.5/2 = 0.25 (divided by 2 because plastic sheet used)
- Now, for the Type/stiffness of the soil: Kf

$$Kf = \frac{1}{1 + \frac{Ac * Ec}{Af * Ef}}$$

$$Ac = 20 * 10 = 200 \text{ m}^2 \quad Af = 2.5 * 200 = 500 \text{ m}^2$$

$$Ef = 40000 \text{ MPa} \quad Af = 2.5 * 200 = 500 \text{ m}^2$$

$$Ec = 30220.3 \text{ MPa} \quad Ef = 40000 \text{ MPa}$$

$$\text{As a result, } Kf = 1/1 + [(200 * 30220.3)/(500 * 40000)] = 0.78$$

$$\text{Thus, } K_{total} = 0.0715 * 0.77 = 0.195$$

- For the stress calculation,
 - $\sigma = E * \alpha * \Delta T * Kr * Kf * \text{creep}$
 - At the 0.4m $\sigma = (0.00001 * 0.25 * 0.78 * 1 * (18 - 9.6) * 17547.7) = 0.98$
- Further calculations is as follows:

strcuture	Hieight	Concrete Lift	Time (days)	0	1	2	3	4		
			Height	Δ Temp (C)	16.386	Δ Temp (C)	3.641	Δ Temp (C) (1st)	4.005	Δ Temp (C) (1st)
Concrete Block	8	Second Lift	8					-0.28849		-0.35106
	7.8		7.8					0.274199		0.143471
	7.6		7.6					0.274199		0.485838
	7.4		7.4					0.274199		0.485838
	7.2		7.2					0.274199		0.485838
	7		7					0.274199		0.485838
	6.8		6.8					0.274199		0.485838
	6.6		6.6					0.274199		0.485838
	6.4		6.4					0.274199		0.485838
	6.2		6.2					0.274199		0.485838
	6		6					0.274199		0.485838
	5.8		5.8					0.274199		0.485838
	5.6		5.6					0.274199		0.485838
	5.4		5.4					0.274199		0.485838
	5.2		5.2					0.274199		0.485838
	5		5					0.274199		0.485838
	4.8		4.8					0.274199		0.485838
	4.6		4.6					0.274199		0.485838
	4.4	4.4					0.274199		0.485838	
	4.2	4.2					0.274199		0.480132	
	4	4	First Lift	4	-0.288485104		-0.351060012		0.264821	0.386932
	3.8	3.8		0.274198564		0.14347054		0.192922	0.455405	
	3.6	3.6		0.274198564		0.485837845		0.396113	0.49725	
	3.4	3.4		0.274198564		0.485837845		0.536784	0.620883	
	3.2	3.2		0.274198564		0.485837845		0.536784	0.706475	
	3	3		0.274198564		0.485837845		0.536784	0.706475	
	2.8	2.8		0.274198564		0.485837845		0.536784	0.706475	
	2.6	2.6		0.274198564		0.485837845		0.536784	0.706475	
	2.4	2.4		0.274198564		0.485837845		1.154905	0.706475	
	2.2	2.2		0.274198564		0.485837845		1.154905	0.706475	
	2	2		0.274198564		0.485837845		1.154905	0.706475	
	1.8	1.8		0.274198564		0.485837845		1.154905	0.706475	
	1.6	1.6		0.274198564		0.485837845		1.154905	0.706475	
	1.4	1.4		0.274198564		0.485837845		1.154905	0.706475	
	1.2	1.2		0.274198564		0.485837845		1.154905	0.706475	
	1	1		0.274198564		0.485837845		1.154905	0.706475	
0.8	0.8	0.274198564			0.485837845		1.154905	0.663679		
0.6	0.6	0.274198564			0.485837845		1.084569	0.601862		
0.4	0.4	0.274198564		0.314654192		0.982974	0.412609			
0.2	0.2	-0.00714327		0.067388916		0.74227	0.177707			
Soil	0.2	Soil	-0.2			-0.179876359		0.501566	-0.05719	
	0.4	Soil	-0.4			-0.351060012		0.399971	-0.24645	
	0.6	Soil	-0.6					0.329635	-0.30826	
	0.8	Soil	-0.8						-0.35106	
	1	Soil	-1							
	1.2	Soil	-1.2							
	1.4	Soil	-1.4							
	1.6	Soil	-1.6							
	1.8	Soil	-1.8							
	2	Soil	-2							
	2.2	Soil	-2.2							
	2.4	Soil	-2.4							
	2.6	Soil	-2.6							
	2.8	Soil	-2.8							
	3	Soil	-3							
	3.2	Soil	-3.2							
	3.4	Soil	-3.4							
	3.6	Soil	-3.6							
	3.8	Soil	-3.8							
	4	Soil	-4							
	4.2	Soil	-4.2							
	4.4	Soil	-4.4							
	4.6	Soil	-4.6							
	4.8	Soil	-4.8							
	5	Soil	-5							
	5.2	Soil	-5.2							
	5.4	Soil	-5.4							
	5.6	Soil	-5.6							
	5.8	Soil	-5.8							
	6	Soil	-6							
	6.2	Soil	-6.2							
	6.4	Soil	-6.4							
	6.6	Soil	-6.6							
	6.8	Soil	-6.8							
	7	Soil	-7							
	7.2	Soil	-7.2							
7.4	Soil	-7.4								
7.6	Soil	-7.6								
7.8	Soil	-7.8								
8	Soil	-8								

➤ **Stress vs Strain:**

- Our compressive strength of concrete is 37.8MPa.
- We have used limestone aggregate in this mix and the elasticity is 19400MPa. (from CEB-FIP model.

Table 5.1-6: Effect of types of aggregate on the modulus of elasticity

Types of aggregate	α_E	$E_{c0} \cdot \alpha_E$ [MPa]
Basalt, dense limestone aggregates	1.2	25800
Quartzite aggregates	1.0	21500
Limestone aggregates	0.9	19400
Sandstone aggregates	0.7	15100

- If we use CEB-FIB model to determine compressive strength values,

2.1.6. Time effects

2.1.6.1. Development of strength with time

The compressive strength of concrete at an age t depends on the type of cement, temperature and curing conditions. For a mean temperature of 20°C and curing in accordance with ISO 2736/2 the relative compressive strength of concrete at various ages $f_{cm}(t)$ may be estimated from eqs (2.1-53) and (2.1-54). To take into account the effect of temperature during curing the actual concrete age should be adjusted according to eq. (2.1-87).

$$f_{cm}(t) = \beta_{cc}(t)f_{cm} \quad (2.1-53)$$

with

$$\beta_{cc}(t) = \exp \left\{ s \left[1 - \left(\frac{28}{t/t_1} \right)^{1/2} \right] \right\} \quad (2.1-54)$$

where

$f_{cm}(t)$ is the mean concrete compressive strength at an age of t days
 f_{cm} is the mean compressive strength after 28 days according to eq. (2.1-1)

$\beta_{cc}(t)$ is a coefficient which depends on the age of concrete t

t is the age of concrete (days) adjusted according to eq. (2.1-87)

$t_1 = 1$ day

s is a coefficient which depends on the type of cement (for cement classification, refer to Appendix d, clause d.4.2.1): $s = 0.20$ for rapid hardening high strength cements RS, 0.25 for normal and rapid hardening cements N and R, and 0.38 for slowly hardening cements SL.

MATERIAL PROPERTIES

Due to the counteracting effects of the parameters influencing the strength under sustained loads, $f_{cm,sus}(t, t_0)$ passes through a minimum. The duration of loading for which this minimum occurs depends on the age of loading and is referred to as the critical period $(t - t_0)_{crit}$. For an age at loading of 28 days, a concrete made of normal cement, type N, $(t - t_0)_{crit} = 2.8$ (days), $f_{c,sus min} = 0.78f_{cm}$. It is generally referred to as sustained load strength of concrete.

There is insufficient experimental basis to give information on the tensile strength of concrete under high sustained tensile stresses.

- We have considered $s = 0.38$ which is SL (slow hardening concrete) because the construction is for bridge and compressive strength is 37.8MPa.

(6) The compressive strength of concrete at an age t depends on the type of cement, temperature and curing conditions. For a mean temperature of 20°C and curing in accordance with EN 12390 the compressive strength of concrete at various ages $f_{cm}(t)$ may be estimated from Expressions (3.1) and (3.2).

$$f_{cm}(t) = \beta_{cc}(t) f_{cm} \quad (3.1)$$

with

$$\beta_{cc}(t) = \exp \left\{ s \left[1 - \left(\frac{28}{t} \right)^{1/2} \right] \right\} \quad (3.2)$$

where:

$f_{cm}(t)$ is the mean concrete compressive strength at an age of t days

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BS EN 1992-1-1:2004 EN 1992-1-1:2004 (E)

f_{cm} is the mean compressive strength at 28 days according to Table 3.1

$\beta_{cc}(t)$ is a coefficient which depends on the age of the concrete t

t is the age of the concrete in days

s is a coefficient which depends on the type of cement:

= 0,20 for cement of strength Classes CEM 42,5 R, CEM 52,5 N and CEM 52,5 R (Class R)

= 0,25 for cement of strength Classes CEM 32,5 R, CEM 42,5 N (Class N)

= 0,38 for cement of strength Classes CEM 32,5 N (Class S)

Note: $\exp\{ \}$ has the same meaning as $e^{ \}$

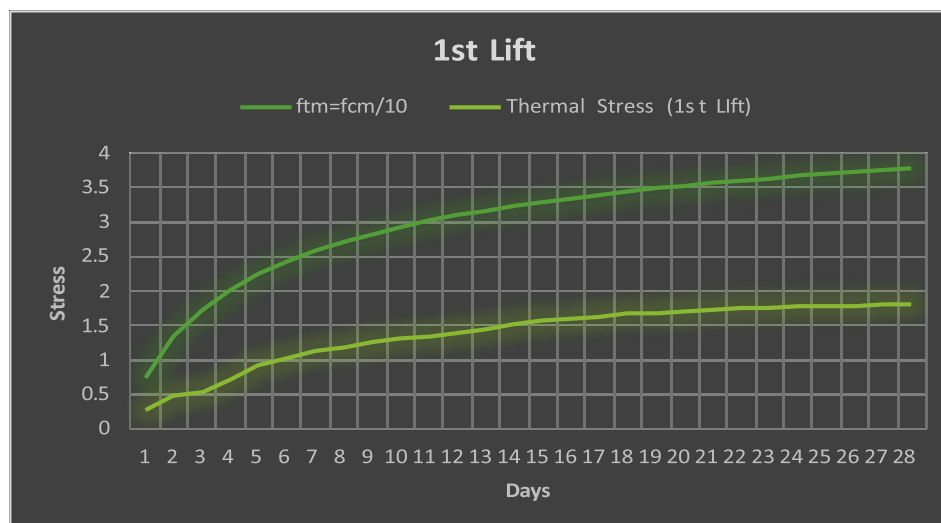
EN 1992-1-1

t	28/t	v	1-v	s	s*1-v	exp(s*1-v)	strength	f _{cm} (Mpa)	power of (2/3)	f _{tm} (Mpa)	t _m =f _{cm} /1	Power of (1/3)	Ec · αE	Ec (Mpa)	Ambient T (C)
				0.38			37.8						19400		
1	28	5.3	-4.3		-1.631	0.196		7.400	3.798	1.435	0.740	0.905		17547.72	18
2	14	3.7	-2.7		-1.042	0.353		13.336	5.624	2.126	1.334	1.101		21353.97	18
3	9.3	3.1	-2.1		-0.781	0.458		17.312	6.692	2.530	1.731	1.201		23294.28	18
4	7	2.6	-1.6		-0.625	0.535		20.225	7.423	2.806	2.023	1.265		24533.82	18
5	5.6	2.4	-1.4		-0.519	0.595		22.490	7.968	3.012	2.249	1.310		25417.38	18
6	4.67	2.2	-1.2		-0.441	0.643		24.323	8.395	3.173	2.432	1.345		26089.94	18
7	4	2.0	-1.0		-0.380	0.684		25.850	8.743	3.305	2.585	1.372		26624.92	18
8	3.5	1.9	-0.9		-0.331	0.718		27.150	9.033	3.415	2.715	1.395		27064.14	18
9	3.1	1.8	-0.8		-0.290	0.748		28.277	9.282	3.508	2.828	1.414		27433.42	18
10	2.8	1.7	-0.7		-0.256	0.774		29.267	9.497	3.590	2.927	1.430		27749.76	18
11	2.5	1.6	-0.6		-0.226	0.798		30.146	9.686	3.661	3.015	1.445		28024.83	18
12	2.3	1.5	-0.5		-0.200	0.818		30.934	9.854	3.725	3.093	1.457		28266.99	18
13	2.2	1.5	-0.5		-0.178	0.837		31.646	10.005	3.782	3.165	1.468		28482.37	18
14	2	1.4	-0.4		-0.157	0.854		32.295	10.141	3.833	3.229	1.478		28675.62	18
15	1.9	1.4	-0.4		-0.139	0.870		32.889	10.265	3.880	3.289	1.487		28850.33	18
16	1.8	1.3	-0.3		-0.123	0.885		33.435	10.379	3.923	3.344	1.495		29009.31	18
17	1.6	1.3	-0.3		-0.108	0.898		33.941	10.483	3.963	3.394	1.503		29154.81	18
18	1.6	1.2	-0.2		-0.094	0.910		34.411	10.579	3.999	3.441	1.510		29288.65	18
19	1.5	1.2	-0.2		-0.081	0.922		34.848	10.669	4.033	3.485	1.516		29412.32	18
20	1.4	1.2	-0.2		-0.070	0.933		35.258	10.752	4.064	3.526	1.522		29527.06	18
21	1.3	1.2	-0.2		-0.059	0.943		35.642	10.830	4.094	3.564	1.528		29633.9	18
22	1.3	1.1	-0.1		-0.049	0.952		36.003	10.903	4.121	3.600	1.533		29733.72	18
23	1.22	1.1	-0.1		-0.039	0.961		36.344	10.972	4.147	3.634	1.537		29827.26	18
24	1.2	1.1	-0.1		-0.030	0.970		36.666	11.037	4.172	3.667	1.542		29915.16	18
25	1.1	1.1	-0.1		-0.022	0.978		36.972	11.098	4.195	3.697	1.546		29997.97	18
26	1.1	1.0	0.0		-0.014	0.986		37.262	11.156	4.217	3.726	1.550		30076.16	18
27	1.0	1.0	0.0		-0.007	0.993		37.537	11.211	4.238	3.754	1.554		30150.15	18
28	1	1.0	0.0		0	1		37.8	11.263	4.257	3.78	1.558		30220.32	18

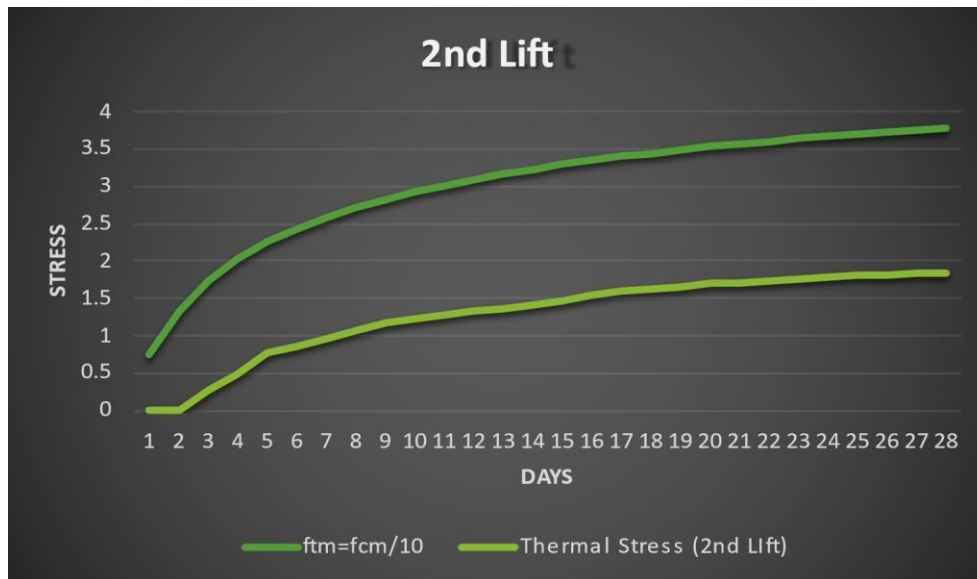
➤ **Thermal Stress for 1st and 2nd lift:**

t (Days)	Thermal Stress (1st Lift)	Thermal Stress (2nd Lift)
1	0.274	-
2	0.486	-
3	0.537	0.274
4	0.706	0.486
5	0.922	0.778
6	1.030	0.863
7	1.127	0.966
8	1.185	1.069
9	1.250	1.162
10	1.303	1.215
11	1.335	1.277
12	1.379	1.328
13	1.433	1.358
14	1.510	1.403
15	1.568	1.457
16	1.604	1.536
17	1.628	1.595
18	1.664	1.631
19	1.676	1.659
20	1.691	1.695
21	1.722	1.710
22	1.741	1.725
23	1.758	1.759
24	1.769	1.780
25	1.783	1.799
26	1.792	1.812
27	1.801	1.826
28	1.807	1.837

➤ **For 1st lift:**



➤ 2nd lift:



- From the graph, it can observe that thermal stress is lower than the concrete strength. So, It is safe to say this it is safe to cast this concrete in single lift and in two lift as well.